

11 - Limits

①

$$1. \lim_{x \rightarrow 0} \left(x + \frac{1}{x}\right)$$

$$= \lim_{x \rightarrow 0} x + \lim_{x \rightarrow 0} \frac{1}{x}$$

$$= \infty$$

DNE

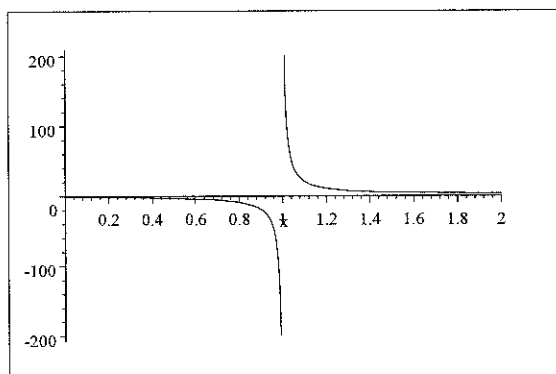
$$2. \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 2x + 1}$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{(x-1)(x-1)}$$

$$= \lim_{x \rightarrow 1} \frac{x+1}{x-1}$$

$$\therefore \lim_{x \rightarrow 1^+} \frac{x+1}{x-1} = +\infty \quad \lim_{x \rightarrow 1^-} \frac{x+1}{x-1} = -\infty$$

DNE



$$3. \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3}$$

$$= \lim_{x \rightarrow 3} x + 3 = 6$$

$$4. \lim_{t \rightarrow 0} \frac{t^2 - t}{t} = \lim_{t \rightarrow 0} \frac{t(t-1)}{t}$$

$$= \lim_{t \rightarrow 0} t - 1 = -1$$

$$5. \lim_{x \rightarrow 2} \frac{\pi}{2} = \frac{\pi}{2}$$

$$6. \lim_{x \rightarrow 0} |x| = 0 \quad |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Since

$$\lim_{x \rightarrow 0^+} x = 0 = \lim_{x \rightarrow 0^-} -(x) = 0$$

$$7. \lim_{x \rightarrow e} \ln x^2 = \lim_{x \rightarrow e} 2 \ln x$$

$$= 2 \ln e = 2$$

$$8. \lim_{x \rightarrow \pi/2} \tan x \text{ DNE} \quad (\text{use graph})$$

since

$$\lim_{x \rightarrow \pi/2^-} \tan x = \infty \neq \lim_{x \rightarrow \pi/2^+} \tan x = -\infty$$

$$9. \lim_{x \rightarrow \pi/2} (1 - \sin x) = 1 - 1 = 0$$

$$10. \lim_{u \rightarrow 2} (u^2 + 4u + 1) = 8 + 8 + 1 = 17$$

$$11. \lim_{x \rightarrow \infty} \left(x + \frac{1}{x}\right)$$

$$= \lim_{x \rightarrow \infty} x + \lim_{x \rightarrow \infty} \frac{1}{x} = \infty + 0 = \infty$$

$$12. \lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{x^3 + 2x^2 + 1}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3}}{1 + \frac{2}{x} + \frac{1}{x^2}} = \frac{0}{1} = 0$$

$$13. \lim_{x \rightarrow 0^+} \ln x = -\infty$$

$$14. \lim_{x \rightarrow 3^-} \frac{-1}{\sqrt{3-x}} = \frac{-1}{\sqrt{3-x}}$$

$$15. \lim_{x \rightarrow 1^+} \frac{x^2 - 3x + 2}{x^2 - 2x + 1}$$

$$= \lim_{x \rightarrow 1^+} \frac{(x-2)(x-1)}{(x-1)(x-1)} = -\infty$$